

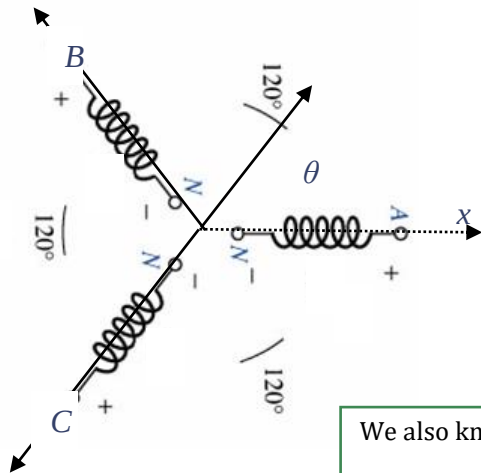
Lecture-5: Applications Induction Machine

Energy Conversion

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Mathematical Interpretation of Rotating M. Field

3 phase AC currents & 120 degree apart in space



$$i_a(t) = I_m \cos(\omega t) \quad i_b(t) = I_m \cos(\omega t - 120^\circ)$$

$$i_c(t) = I_m \cos(\omega t - 240^\circ)$$

We know that $B=Ki$

$$B_a = Ki_a(t) \cos(\theta) \quad B_b = Ki_b(t) \cos(\theta - 120^\circ)$$

$$B_c = Ki_c(t) \cos(\theta - 240^\circ)$$

$$B_{\text{total}} = KI_m \cos(\omega t) \cos(\theta) + KI_m \cos(\omega t - 120^\circ) \cos(\theta - 120^\circ)$$

$$+ KI_m \cos(\omega t - 240^\circ) \cos(\theta - 240^\circ)$$

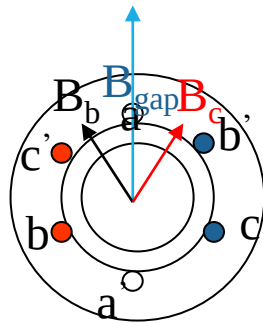
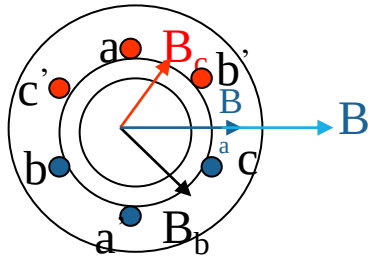
We also know that $\cos(x) \cos(y) = (1/2)[\cos(x - y) + \cos(x + y)]$

$$B_{\text{total}} = \frac{3}{2} KI_m \cos(\omega t - \theta) + \frac{1}{2} KI_m [\cos(\omega t + \theta)$$

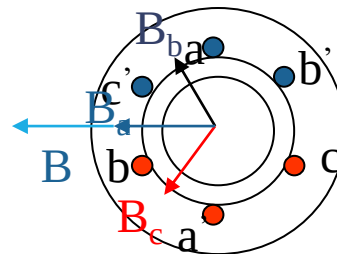
$$+ \cos(\omega t + \theta - 240^\circ) + \cos(\omega t + \theta - 480^\circ)]$$

$$B_{\text{total}} = B_a + B_b + B_c = \frac{3}{2} B_m \cos(\omega t - \theta)$$

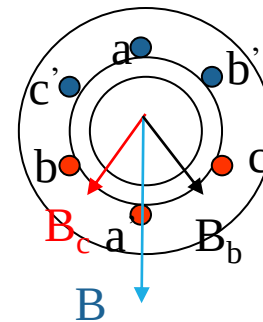
Interpretation of Rotating Magnetic Field



$t = t_1, \omega t = 90$



$t = t_2, \omega t = 180$



$t = t_3, \omega t = 270$

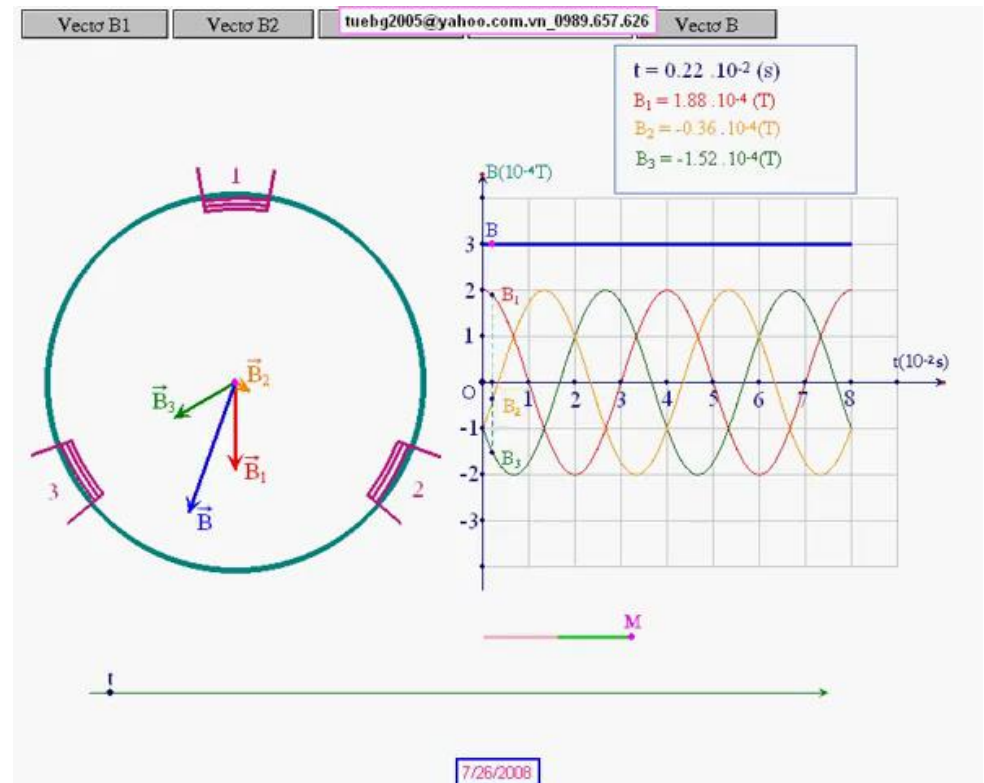
$$B_{\text{gap}} = B_m \cos(\omega t - \theta)$$

$$B_a(t) = B_m \cos(\omega t)$$

$$B_b(t) = B_m \cos(\omega t - 120^\circ)$$

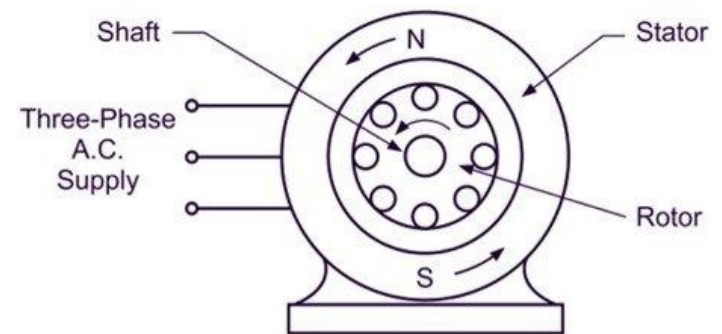
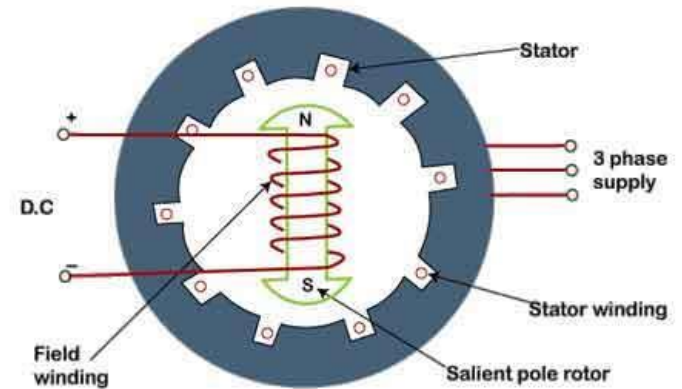
$$B_c(t) = B_m \cos(\omega t - 240^\circ)$$

- Interpretation of Rotating Magnetic Field

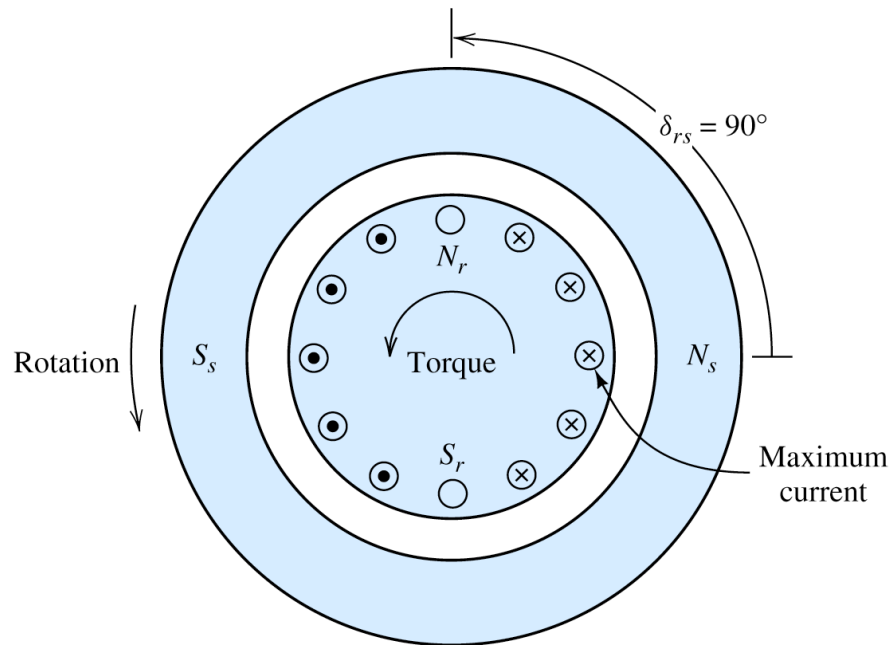


Synchronous and Asynchronous (Induction) Machine

- Synchronous machines
- Asynchronous (Induction) machines



Slip and Slip Ratio



Cross section of a squirrel-cage induction motor. The rotating stator field induces currents in the conducting bars which in turn set up magnetic poles on the rotor. Torque is produced because the rotor poles are attracted to the stator poles.

Synchronous Speed

$$\omega_s = \frac{\omega}{P/2} \quad n_s = \frac{120 f}{P}$$

$$S = \frac{\omega_s - \omega_m}{\omega_s} = \frac{n_s - n_m}{n_s}$$

$$\omega_{\text{slip}} = S\omega$$

$$n_m = (1-s)n_{sync} \text{ and } \omega_m = (1-s)\omega_{sync}$$

Induction Motor

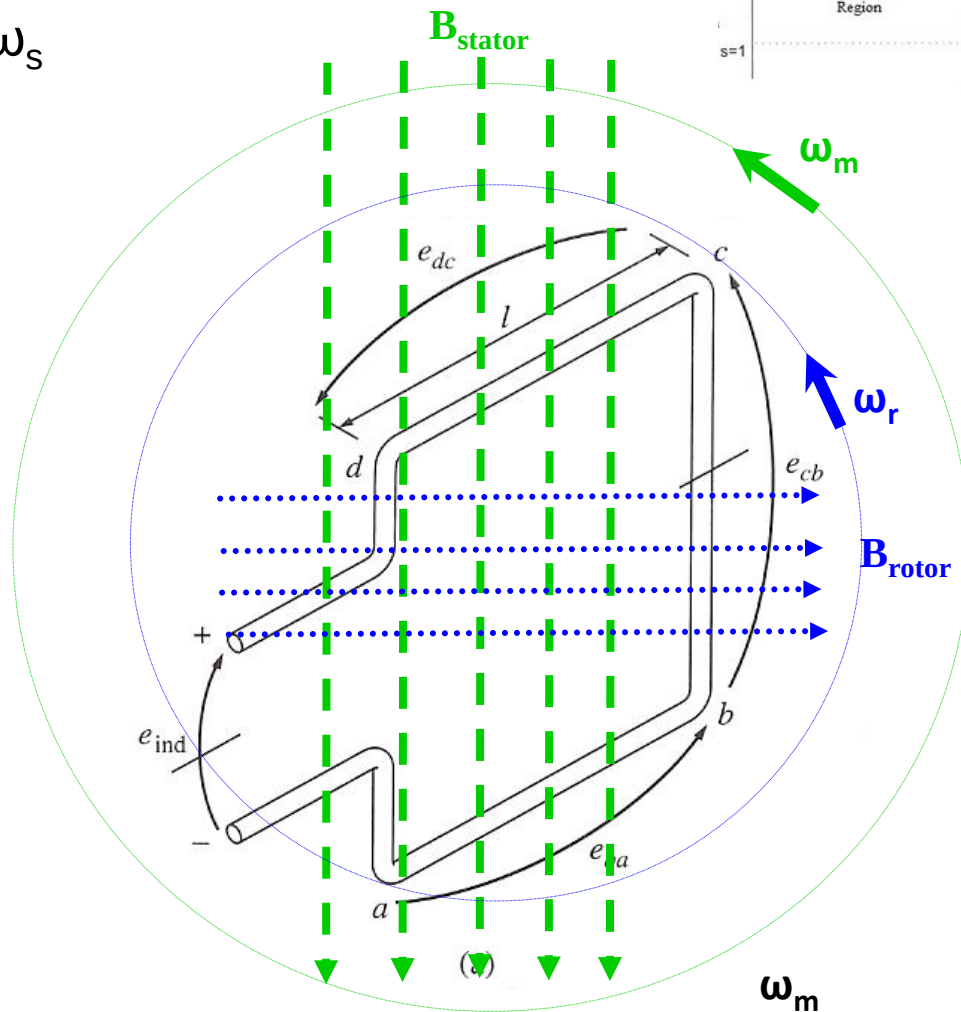
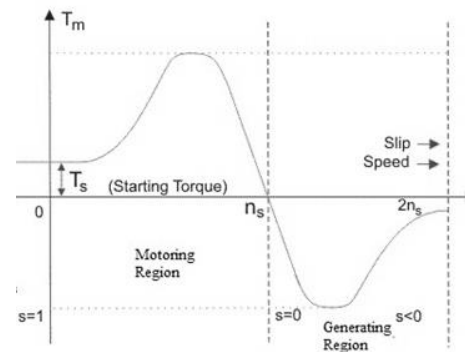
- Rotating speed ω_m , ω_r , ω_s

- Motor

Slip ratio ≥ 0
 Synchronous machine
 Induction machine

- Generator

Slip ratio < 0
 Induction machine



Induced Voltage in AC Machines

- The total voltage induced in the coil will be sum of the voltages induced in each of its four sides.

- *Segment ab* : $e_{ba} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} = vBl$ (directed out of the page)

$$= vB_M l \cos(\omega_m t - 180^\circ)$$

- *Segment bc and da* : The induced voltage is zero, since

$$\text{So } (\mathbf{v} \times \mathbf{B}) \perp \mathbf{l} \quad \cdot \quad e_{cb} = e_{ad} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} = 0$$

- *Segment cd* : $e_{dc} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} = vBl$ (directed out of the page)

- Therefore, $= v(B_M \cos \omega_m t)l = vB_M l \cos \omega_m t$

$$e_{ind} = e_{ba} + e_{dc} = vB_M l \cos \omega_m t + vB_M l \cos \omega_m t$$

$$= 2vB_M l \cos \omega_m t = 2(r\omega_m)B_M l \cos \omega_m t = 2r l B_M \omega_m \cos \omega_m t$$

Induced Voltage in AC Machines

- Finally, the flux passing through the coil can be expressed as $\phi = 2rlB_M$, while $\omega_m = \omega_e = \omega$ for a two-pole stator, and if the coil in the stator has N_C turns of wire, then the total induced voltage of the coil will be

$$e_{ind} = N_C \phi \omega \cos \omega t \text{ (N turns of wires in the coil)}$$

- Notice that the voltage produced in stator of this simple AC machine winding is sinusoidal with an amplitude that depends on the flux Φ in the machine, the angular velocity ω of the rotor, and a constant depending on the construction of the machine.
- If the reference direction for α had been rotated by 90° , we would have had a $\sin \omega t$ term instead of $\cos \omega t$ in the above equation.

Induced Voltage in AC Machines

- The induced voltage in a 3Φ set of coils
 - If three coils, each of N_C turns, are placed around the rotor magnetic field, then the voltages induced in each of them will be the same in magnitude but will differ in phase by 120° .
 - The resulting voltages in each of the three coils are

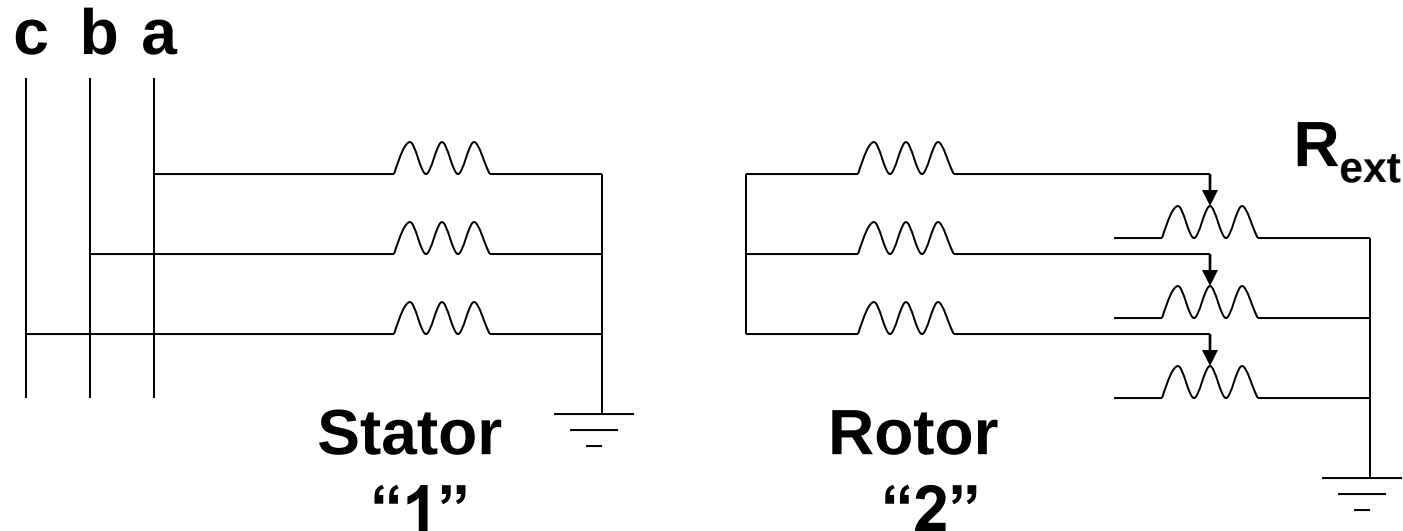
$$e_{aa'} = N_C \phi \omega \cos \omega t$$

$$e_{bb'} = N_C \phi \omega \cos(\omega t - 120^\circ)$$

$$e_{cc'} = N_C \phi \omega \cos(\omega t - 240^\circ)$$

- Therefore, a 3Φ set of currents can generate a uniform rotating magnetic field in a machine stator, and a uniform rotating magnetic field can generate a 3Φ set of voltages in such a stator.

Equivalent Circuit of Induction Motor



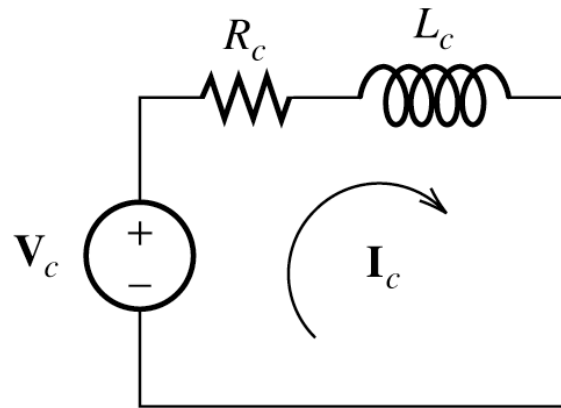
Assumptions:

Both stator and rotor have 3- ϕ Y-connected windings.

The stator is fed from a 3- ϕ source.

The subscript "1" refers to the stator and "2" refers to the rotor.

Induced Voltage in Rotor



Equivalent circuit for a rotor conductor. V_c is the phasor for the induced voltage, R_c is the resistance of the conductor, and L_c is the inductance.

What is the frequency of the induced voltage?

Nature Induction Machine

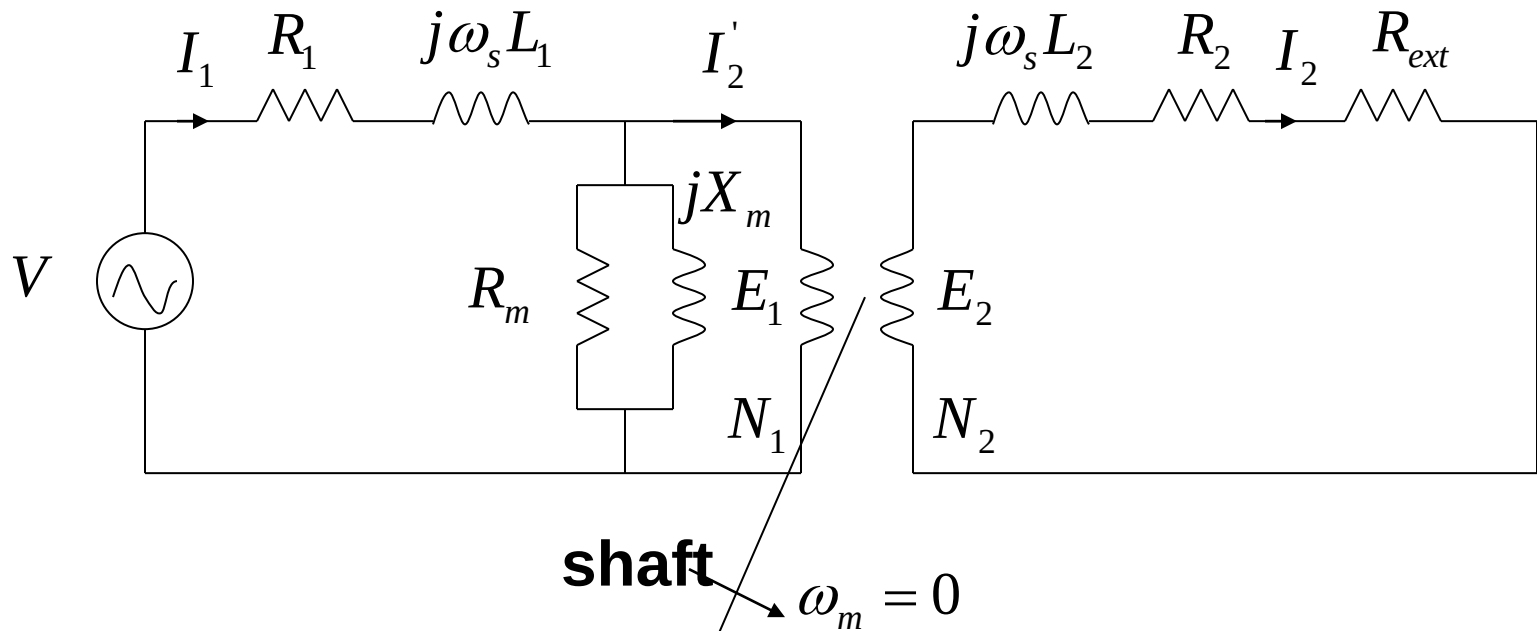
What is IM
when it's not
rotating?

What is IM
when it's
rotating?

a) Standstill condition

Suppose the rotor is at standstill (that is rotor is stationary, $\omega_m=0$). Induction Machine Becomes a Transformer, a narrow air gap is necessary.

Using transformer theory, the equivalent circuit for 1- ϕ to ground is:



a) Standstill condition

Normally X_m and R_m are very large and net Z_m acts as open circuit. Therefore, they are ignored and

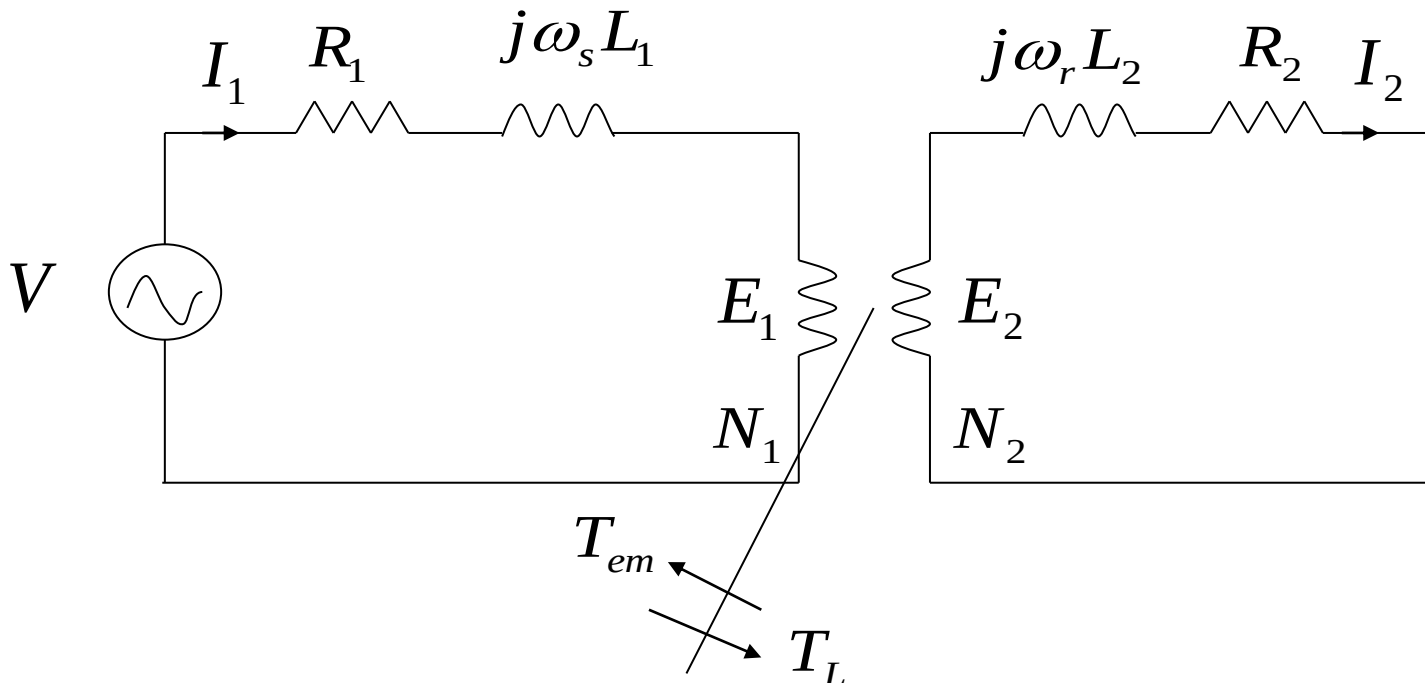
$$I_1 = I_2'$$

$$\left. \begin{aligned} V &= I_1(R_1 + j\omega_s L_1) + E_1 \\ E_2 &= I_2(R_2 + R_{ext} + j\omega_s L_2) \end{aligned} \right\} \text{Standstill}$$

b) Rotor is free to rotate

Suppose rotor speed is ω_m .

And $R_2 = R_{2,\text{rotor}} + R_{\text{ext}}$



b) Rotor is free to rotate

$$V = I_1(R_1 + jX_1) + E_1$$

$$E_2 = I_2(R_2 + j\omega_r L_2)$$

Where

R_1 : stator resistance/ ϕ

$X_1 = \omega_s L_1$: stator reactance/ ϕ

I_1 : stator current (line)

E_1 : induced emf/ ϕ

b) Rotor is free to rotate

Relationship between I_1 and I_2 , and E_1 and E_2

Define

$$a = \frac{N_1}{N_2} \quad I_1 = \frac{I_2}{a}$$

We know that

$$e_1 = N_1 \frac{d\Phi}{dt} = \omega N_1 \Phi_{\max} \cos(\omega t + \theta_1)$$

$$e_2 = N_2 \frac{d\Phi}{dt} = s \omega N_2 \Phi_{\max} \cos(\omega_r t + \theta_2)$$

b) Rotor is free to rotate

$$E_1 = \frac{\omega N_1 \Phi_{\max}}{\sqrt{2}}$$

$$E_2 = \frac{s \omega N_2 \Phi_{\max}}{\sqrt{2}}$$



$$E_2 = s N_2 \frac{E_1}{N_1} = s \frac{E_1}{a}$$

So

$$I_2 = a I_1$$

$$E_2 = s \frac{E_1}{a}$$

Reduced circuit model

$$E_2 = I_2(R_2 + j\omega_r L_2)$$

 $s \frac{E_1}{a} = a I_1(R_2 + j\omega_r L_2)$

Multiply by a

 $sE_1 = I_1(a^2 R_2 + jsa^2 X_2)$

Divided by s

 $E_1 = I_1(a^2 \frac{R_2}{s} + ja^2 X_2)$

Reduced circuit model

But from the stator equation

$$V = I_1(R_1 + jX_1) + E_1$$

So

$$V = I_1(R_1 + jX_1) + I_1(a^2 \frac{R_2}{s} + ja^2 X_2)$$

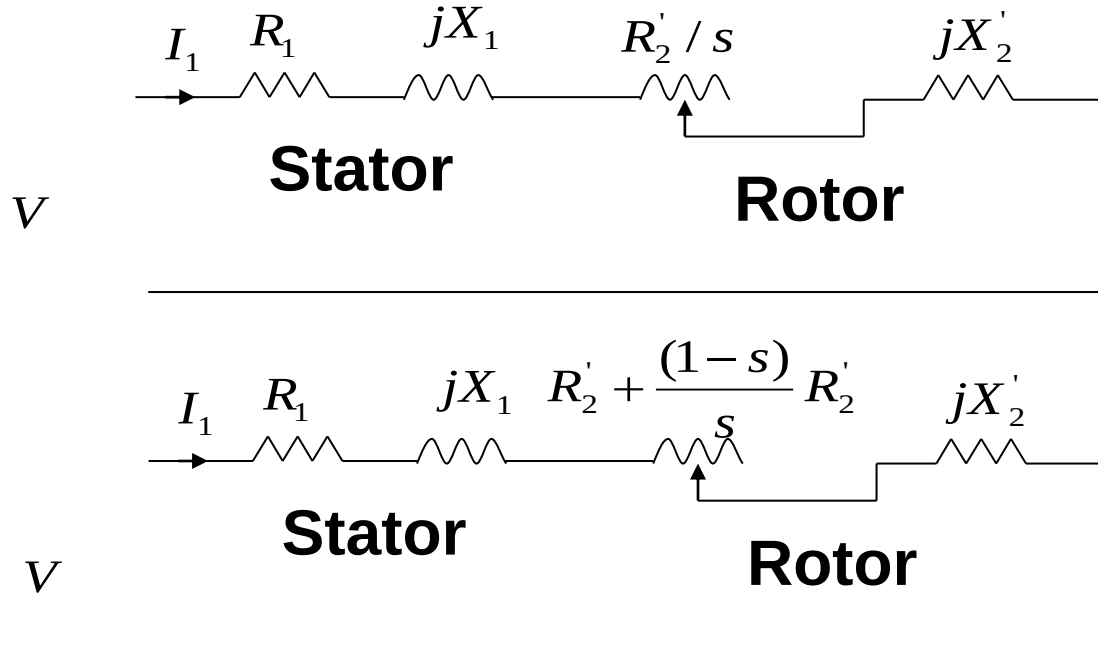
Let's define

$$R_2' = a^2 R_2 \quad X_2' = a^2 X_2$$

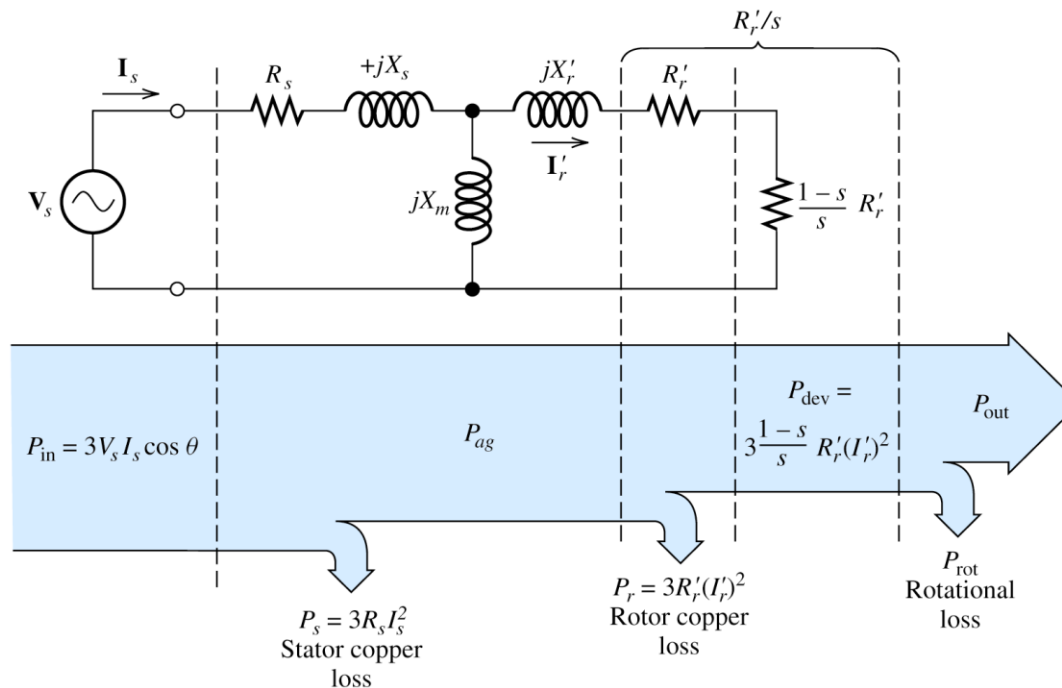
Reduced circuit model

$$V = I_1 \left[\left(R_1 + \frac{R_2'}{s} \right) + j(X_1 + X_2') \right]$$

The induction machine equivalent circuit is:



Induction Motor Equivalent Circuit



Equivalent circuit for one phase of an induction motor and the associated power-flow diagram. V_s is the rms phase voltage and I_s is the rms phase current.

Course notes of ECE 3113, Energy Conversion